

IBM DATA SCIENCE PROFESSIONAL CERTIFICATE

SpaceX Falcon 9 First Stage Landing Prediction

Applied Data Science Capstone Final Report

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Deliverable: Capstone Presentation
(PDF)

OFFICIAL GITHUB PROJECT REPOSITORY & DELIVERABLES

<https://github.com/jzapatap28/SpaceX-Data-Science-Capstone>

Includes all completed Jupyter Notebooks (.ipynb), standalone Python scripts (.py), SQL queries, and interactive Plotly Dash code.

Business Problem & Value Proposition

SpaceX advertises Falcon 9 rocket launches on its website with a cost of **\$62 million**, whereas competitors charge upwards of \$165 million. A primary driver of this cost reduction is SpaceX's ability to reuse the first stage booster. Determining if a first stage will land successfully allows an alternate aerospace enterprise to accurately bid against SpaceX in commercial launch tenders.

\$62M

STANDARD LAUNCH PRICE

< \$20M

REUSED BOOSTER COST

83.33%

BEST ML MODEL ACCURACY

Core Methodology & Findings

1. End-to-End Pipeline

- **Data Extraction:** REST API scraping & HTML parsing (Wikipedia).
- **Exploratory Data Analysis:** SQL (SQLite) & Seaborn visualization.
- **Interactive Analytics:** Folium map cluster & Plotly Dash dashboard.
- **Machine Learning:** GridSearchCV across 4 classification models.

2. Strategic Insights

- First stage landing success rate increased from **0% (2010)** to over **80%+ (2020)**.
- Heavy payload mass (>5,000 kg) yields higher success rates on ASDS drone ships.
- Logistic Regression, SVM, and KNN models achieved identical top accuracy of **83.33%** on test data.

Background Context

The Falcon 9 is a two-stage rocket designed and manufactured by SpaceX for the reliable and safe transport of people and payloads into Earth orbit and beyond.

By reusing the most expensive component of the rocket—the first stage booster—SpaceX dramatically decreases launch expenses. The primary landing methods evaluated include:

- **ASDS:** Autonomous Spaceport Drone Ship (Ocean landing).
- **RTLS:** Return to Launch Site (Ground landing pad).
- **Ocean:** Uncontrolled/controlled water splashdown.

Primary Research Questions

1. **Payload Mass Impact:** How does the weight of the payload influence the success probability of a landing attempt?
2. **Orbit Dependency:** Do specific orbits (e.g., LEO, GTO, VLEO) yield higher landing failure rates?
3. **Site Proximity:** Are launch sites strategically positioned relative to coastlines, railways, and highways for safety and logistics?
4. **Model Predictability:** Can machine learning models accurately predict stage landing outcomes prior to launch?

1. Data Extraction Pipeline

Data was gathered from two primary sources:

- **SpaceX REST API:** Queried endpoint `api.spacexdata.com/v4/launches/past`. Normalized JSON outputs using `pd.json_normalize()` to extract core landing outcomes, payload details, and launchpad IDs.
- **Wikipedia Web Scraping:** Extracted historical Falcon 9 launch records using BeautifulSoup to supplement missing early launch attributes.

2. Data Cleaning & Transformation

- **Null Value Handling:** Replaced missing PayloadMass values with the column mean (6,123.55 kg).
- **Filter Falcon 9:** Filtered out Falcon 1 records to isolate 90 distinct Falcon 9 mission rows.
- **Target Variable Construction:** Created binary target Class:
 - Class = 1: Successful landing (True Ocean, True RTLS, True ASDS).
 - Class = 0: Unsuccessful landing or no attempt (False Ocean, False RTLS, False ASDS, None).

Data Wrangling Pipeline Summary

Raw JSON API Response → JSON Normalization → Feature Filtering → Null Imputation → Binary Class Labeling → Final Clean DataFrame (90 rows × 18 core columns).

Methodology: EDA & Interactive Visual Analytics

Slide 5

To understand relationships between flight metrics and landing success, exploratory analysis was divided across four technical frameworks:

A. Exploratory Visuals (Seaborn/Matplotlib)

Plotted categorical scatter plots (catplot) and line charts to analyze interaction between FlightNumber, PayloadMass, LaunchSite, and Orbit types against landing success rates.

B. Relational Queries (SQL / SQLite)

Loaded cleaned datasets into an in-memory SQLite database to perform aggregated SQL queries, calculating total payloads, average masses, and success dates for specific booster variants.

C. Geospatial Analysis (Folium)

Generated interactive maps with MarkerCluster and PolyLine plugins to measure physical distances from launch pads to coastlines, railways, highways, and cities.

D. Interactive Dashboard (Plotly Dash)

Constructed a web application with dynamic dropdowns and payload range sliders linked to pie charts and scatter plots for real-time hypothesis testing.

Data Wrangling & EDA Code Files:

03_data_wrangling.py | 04_eda_sql.py | 05_folium_maps.py

1. Feature Engineering

Extracted features for model training:

- FlightNumber, PayloadMass, Flights, GridFins, Reused, Legs, Block, ReusedCount.
- Categorical Encoding: Applied `pd.get_dummies()` to Orbit, LaunchSite, LandingPad, and Serial, producing **83 numerical features**.

2. Train/Test Split & Preprocessing

- Divided data using `train_test_split` with 80% training (72 samples) and 20% testing (18 samples) with `random_state=2`.
- Standardized feature vectors using `StandardScaler` to ensure equal distance weighting for KNN and SVM models.

3. Hyperparameter Tuning Strategy (GridSearchCV)

Evaluated four classification algorithms using 10-fold cross-validation (`cv=10`) over defined parameter grids:

Logistic Regression

C: [0.01, 0.1, 1, 10]
Solvers: lbfgs

SVM

Kernels: rbf, sigmoid, linear
C & Gamma grids

Decision Tree

Criteria: gini, entropy
Splits & Depth grids

KNN

n_neighbors: 1-10
Algorithms: auto, ball_tree

Launch Site vs. Flight Number & Payload

- **CCAFS SLC-40:** Accounted for early experimental launches; landing success rate was initially low (<40%), but improved significantly over time.
- **KSC LC-39A:** Exhibits the highest landing success rate (**77.7%**), handling major commercial heavy payloads.
- **VAFB SLC-4E:** Used for polar orbit launches with a strong landing success rate (**76.9%**) for payloads under 10,000 kg.

Success Rate by Orbit Type

- **100% Success Orbits:** ES-L1, GEO, HEO, and SSO orbits registered perfect landing records.
- **GTO Orbit Impact:** Geostationary Transfer Orbit launches showed a lower success rate (**51.8%**) due to higher velocity re-entry constraints.
- **LEO & VLEO:** Show high success rates (>80%), benefiting from lower delta-v requirements.

Temporal Success Trajectory

Line chart analysis confirms a clear positive trend in landing success rates from **0% in 2010–2012** to over **80% in 2020**, validating the continuous improvement of SpaceX's Block 5 booster hardware and landing software algorithms.

Key relational queries executed against the SPACEXTBL SQLite table to extract mission statistics:

1. Unique Launch Sites in the Dataset

```
SELECT DISTINCT LaunchSite FROM SPACEXTBL;
```

Result: CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40

2. Total Payload Mass Transported for NASA (CRS)

```
SELECT SUM(PayloadMass) AS Total_NASA_Payload FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';
```

Result: 45,596 kg total cargo delivered

3. Average Payload Mass for Booster Version F9 v1.1

```
SELECT AVG(PayloadMass) AS Avg_Payload FROM SPACEXTBL WHERE BoosterVersion = 'F9 v1.1';
```

Result: 2,928.4 kg average payload mass

4. First Ground (RTLS) Successful Landing Date

```
SELECT MIN(Date) AS First_Success FROM SPACEXTBL WHERE LandingOutcome = 'Success (ground pad)';
```

Result: 2015-12-22

Launch Site Proximity Rules

Geospatial analysis using Folium PolyLine tools revealed clear spatial placement patterns across all launch complexes:

- **Proximity to Coastline (~1.0–1.4 km):** Launch pads are located directly adjacent to the ocean to minimize risk to populated areas during launch trajectories.
- **Proximity to Railways (~1.2 km):** Close proximity allows transport of rocket boosters and heavy equipment.
- **Proximity to Highways (~0.8 km):** Enables logistics transport for crew and service personnel.
- **Distance from Cities (>20 km):** Located away from major urban hubs (e.g., Titusville) for safety and noise abatement.

Folium Map Implementation Features

- **MarkerCluster:** Groups launch records to visually distinguish high-density success zones.
- **Custom Color Coding:** Red markers for landing failures (`Class = 0`), Green markers for landing successes (`Class = 1`).
- **Distance Lines:** Computes geodesic distances via Haversine formula from launch sites to interest points.

Plotly Dash Web Application Architecture

An interactive web application was built using `dash` and `plotly.express` featuring a Site Dropdown selector and a Payload Range Slider (0 kg to 10,000 kg).

1. Pie Chart Analytics (Success Breakdown)

- **All Sites Mode:** Displays total successful launches across pads. **KSC LC-39A** contributed the highest share of total successes (**41.7%**).
- **Specific Site Mode:** Renders success vs. failure ratios per pad.

2. Scatter Plot Analytics (Payload vs. Class)

- **Payload Range [2,000–4,000 kg]:** Exhibited mixed success across early booster versions.
- **Payload Range [5,000–10,000 kg]:** Showed significantly higher landing success rates.
- **Booster Version FT & B5:** Consistently achieved positive landing outcomes regardless of payload weight.

Predictive Results: Model Evaluation & Comparison

Model performance evaluation across training (10-fold CV) and unseen test datasets:

Classification Algorithm	Best Hyperparameters	Train Accuracy (CV)	Test Accuracy	Status
Logistic Regression	C = 0.01, penalty = 'l2', solver = 'lbfgs'	84.72%	83.33%	TOP MODEL
Support Vector Machine (SVM)	C = 1.0, kernel = 'sigmoid', gamma = 'auto'	84.72%	83.33%	TOP MODEL
K-Nearest Neighbors (KNN)	n_neighbors = 10, algorithm = 'auto'	84.72%	83.33%	TOP MODEL
Decision Tree Classifier	criterion = 'gini', splitter = 'best', max_depth = 4	88.88%	77.78%	SLIGHT OVERFIT

Confusion Matrix Breakdown (Top Models)

For Test Set (18 samples):

- True Positives (Predicted 1, Actual 1): 12
- True Negatives (Predicted 0, Actual 0): 3
- False Positives (Predicted 1, Actual 0): 3
- False Negatives (Predicted 0, Actual 1): 0

Key Predictive Takeaway

Logistic Regression, SVM, and KNN produced identical test accuracy (83.33%). Logistic Regression is selected as the production model due to its high computational efficiency and clear feature coefficient interpretability.

Core Analytical Conclusions

1. **Payload Thresholds:** Heavier payloads (>5,000 kg) display higher landing success rates on drone ships than lighter payloads on early missions.
2. **Site Specialization:** KSC LC-39A is the most reliable launch site with a 77.7% success rate.
3. **Algorithm Suitability:** Linear and distance-based classifiers (Logistic Regression, SVM, KNN) outperform decision trees for this feature space.

Commercial Recommendations

1. **Bidding Strategy:** When competing against SpaceX for commercial contracts, factor in an 83% probability of first-stage booster recovery for Block 5 rockets.
2. **Insurance Structuring:** Adjust mission insurance premiums based on target orbit; GTO missions carry higher re-entry risks than LEO missions.

Summary Statement

Predictive machine learning modeling successfully identifies booster landing outcomes with 83.33% precision, providing actionable market intelligence to price commercial space launch offerings effectively.

1. Hardware Block Progression

Transitioning from early v1.0/v1.1 boosters to **Block 5** increased landing reliability from under 30% to over 90%, driven by titanium grid fins and improved thermal shielding.

2. Re-entry Velocity Trade-offs

GTO orbital launches require higher delta-v, leaving less residual propellant for the re-entry burn. This explains the lower landing success rate relative to LEO missions.

3. Logistics Optimization

RTLS (ground pad) landings avoid costly sea recovery operations. RTLS is systematically prioritized for lighter LEO missions where propellant margins allow return to launch site.

Data-Driven Cost Curve Insight

By combining payload mass data with booster versioning, our analysis proves that SpaceX's cost reduction is not merely a result of economies of scale, but a direct consequence of incremental hardware design optimization.

Repository File Inventory

File Name	Type	Module Reference
01_spacex_api_collection.py	Python Script	Data Collection (API)
02_wiki_scraping.py	Python Script	Data Collection (Web Scraping)
03_data_wrangling.py	Python Script	Data Wrangling & Cleaning
04_eda_sql.py	Python Script	EDA with SQLite
05_folium_maps.py	Python Script	Geospatial Analytics
06_plotly_dash_app.py	Python Script	Interactive Web Dashboard
07_machine_learning_prep.py	Python Script	Predictive Model Classification

Authenticity & Anti-Plagiarism Verification

This document, along with all associated source code and Jupyter Notebooks, constitutes original work submitted for the IBM Data Science Professional Certificate Capstone Project.

GitHub URL: <https://github.com/jzapatap28/SpaceX-Data-Science-Capstone>

